

What the Solana code does

The marketplace works without Solana. The Solana program is a shared, public notebook where each order's key facts are written once and cannot be quietly edited. It does not hold money in this version.

Basic Solana ideas used

Account	A piece of on-chain storage. Programs hold no data themselves; data lives in accounts the program owns.
PDA (Program Derived Address)	An account address computed from seeds + the program id. Ours: seeds ["settlement", sha256(order id)], so every order has exactly one predictable record and no one holds its private key.
Instruction	One call into the program. We have three: <code>commit_order</code> , <code>record_attestation</code> , <code>settle</code> .
Signer	An account that signed the transaction. Only the marketplace authority key may change a record (Anchor's <code>has_one = authority</code>).
Transaction & fee	Instructions are sent in a signed transaction. The authority pays the fee and the one-time rent for the new account.
Hash	Only 32-byte fingerprints go on chain. Names, photos and locations stay in the database, but anyone can check them against the hash.

Life of an order on chain

1. Order accepted	<code>commit_order</code> creates the order's PDA: order hash, amount (minor units), status COMMITTED.
2. Each delivery step	<code>record_attestation</code> : <code>chain = sha256(chain + new hash)</code> . Order and content of the evidence become tamper-evident; count goes up.
3. Completed / refunded	<code>settle</code> sets status COMPLETED (1) or REFUNDED (2). After this the program rejects every further change.

What the program refuses (its validation)

- PDA seeds + bump are re-checked on every call, so a fake account is rejected.
- `has_one = authority`: wrong signer fails.
- require! `status == COMMITTED`: no changes after settlement, no double settle.
- Outcome must be 1 or 2.
- Creating the same order twice fails (the account already exists).

Where the code lives

blockchain/solana/programs/marketplace-settlement/src/lib.rs - the Anchor program (Rust).
api/app/payments/solana_adapter.py - Python client that builds and sends the transactions.
api/app/payments/service.py - listens to order events; failed chain calls are saved as FAILED and retried, never blocking an order.